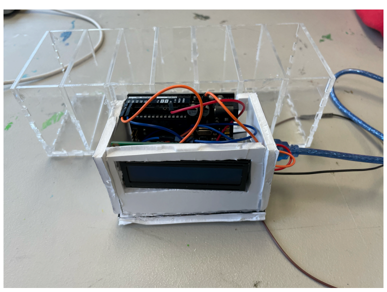
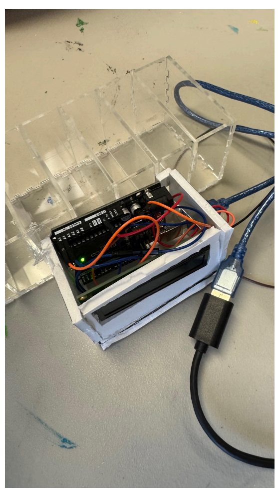
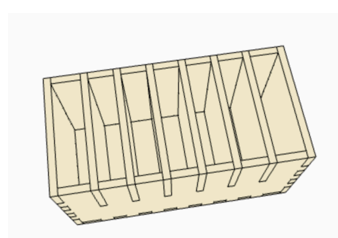
Medication adherence is a critical factor in managing chronic illnesses, yet many patients forget to take their prescribed doses on time. This can lead to serious health complications, reduced treatment effectiveness and increased medical costs. Traditional pillboxes rely on manual tracking, which can be ineffective with memory challenges or busy schedules.

This project aims to develop a smart pillbox using an Arduino and an MC-38 magnetic sensor to monitor whether the pillbox has been opened within a specified timeframe. If the pillbox remains unopened past the scheduled medication time, the system will trigger an alert (such as a buzzer, LED, or mobile notification) to remind the user to take their medication. By automating adherence tracking, this smart pillbox can help patients manage their medication schedules more effectively, improving health outcomes and reducing the risk of missed doses.

Our first prototype was built from Styrofoam, and we made it smaller without thinking of adding the Arduino yet. The first prototype we made had a lid but was too small for the wiring. We used mostly the Styrofoam and hot glue to make it all stay together. Our box had 7 slots in a row, and the lid did not fit properly. We knew when creating the next prototype, we had to make it bigger for the Arduino to fit. The only issue is it is a pill box so it couldn't be too big to where it is now not reasonable. We didn't get to exactly test it with Arduino, we had to move straight into doing the plexiglass prototype. With the limited amount of time our plexiglass pill box didn't have a lid. The lid is such a main function and part of our design. This was supposed to alert the stakeholder to take their meds if it remains unopened. These changes to our first prototype were necessary for the operation of the pill box. We created our final prototype to be 3D printed and designed it online first. After it was printed we used the styrofoam to create a small box to glue on the side holding the wires and Arduino.

We are testing the model using the Arduino for it to send a code to the stakeholder's phone. This covers one of our highest stakeholders of being big enough to hold all medications and alerting the person to take their medications. Our feedback from experts included not only having 7 spots for days of the week but also am and pm slots since some people take medications in the morning and at night, different times of the day. The IT workers helped give us feedback on things we had not thought about previously. We struggled with connecting the wires to make it read and notify the phone. This is something we would need to address and fix if we had time to fix the lid of the pill box.